



ARCHER2 CSE Quarterly Report

April–June 2026

EPCC

The University of Edinburgh



Document Information and Version History

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CSE Quarterly Report

This report covers ARCHER2 CSE activities during the period 1 April—30 June 2026.

Executive Summary

- The CSE team has fulfilled all its service level agreements.
- The AMD tools and drivers on the GPU Development Service have been upgraded to a more current version.
- The PowerSched evaluation has concluded, and a full-system trial is recommended.
- There was a strong CSE presence at both the Cray User Group meeting and the International Supercomputing Conference (ISC).
- A total of 22.5 days of training has been delivered as part of the ARCHER2 CSE training programme.
- A very successful session on [Research software development funding calls](#) was presented at the [Durham HPC Days](#), included a talk on the eCSE programme and the Software Sustainability Institute (SSI) Research Software Maintenance Fund (RSMF).
- EPCC was delighted to have passed our annual ISO external audits for Quality Service Delivery, Information Security and Business Continuity and Disaster Recovery

Forward Look

- ARCHER2 Exit planning is progressing.
- The ARCHER2 CSE team are exhibiting at the Royal Society Summer Science Exhibition, in London, during 30th June—5th July.

ARCHER2 Centralised CSE Team

As in previous period, the CSE team has been active in the scientific-computing community, promoting the ARCHER2 service and gathering feedback from researchers (mostly towards future services, given the age of ARCHER2).

The CSE team was represented by Juan Rodriguez Herrera and David Henty at the Cray User Group meeting which, this year, was held during 26th—30th April in Nice, France.

Juan contributed to various activities:

- Speaker at the HPC System Test Workshop (CUG 2026 satellite workshop).
- Panellist on "Read the Docs: Best Practices for Site Documentation" (BoF), part of PEAD (Programming Environments, Applications and Documentation), which is a Special Interest Group that meets annually at CUG.
- Panellist on "Operational Models for PyTorch on HPE Cray Systems".

David gave a talk on "Performance of a Hybrid MPI/OpenMP Sparse Matrix Solver on Slingshot 10 and 11" which compared the performance of benchmarks on ARCHER2 and the new Cirrus National Compute Resource system.

The 2026 run of the International Supercomputing Conference (ISC) was held close to the end of the period (during 22nd–26th June, in Hamburg, Germany) and included significant representation from CSE (Eleanor Broadway, James Richings, and Andy Turner) – both for the technical programme and various Women in HPC activities (described later).

James Richings gave a talk about sustainable supercomputing on GPU frequency optimisation, based on work undertaken on the ARCHER2 GPU Development System. Andy Turner gave talks at the 3rd International Workshop on Readiness of HPC Extreme-scaling Applications and at the 5th International Workshop on Sustainable Supercomputing.

EPCC ran a booth showcasing HPC services including both ARCHER2 and the Cirrus NCR. Further, all three participants were WHPC mentors for the poster programme.

In other news, a paper describing seamless QM/MM simulations using GROMACS and CP2K, to which both CSE (Arno Proeme) and the HPE Centre of Excellence (Holly Judge) contributed, has been accepted for publication in the Journal of Chemical Information and Modelling (JCIM).

Closer to home, the IT world has been impacted by the announcement of several, newly detected, cybersecurity vulnerabilities. CSE supported SP and HPE to test and finalise patched node images for ARCHER2, which were then deployed using the new, lightweight upgrade mechanism (described in a previous quarterly report). A similar process was used for the new NCR, Cirrus.

Continual Service Improvement (CSI) Projects

PowerSched

CSE completed its evaluation of PowerSched, the tool being developed by HPE and researchers at HLRS (Germany), to efficiently manage the power draw of an HPC service in response to operating conditions such as power costs and cooling performance.

High-level conclusions are:

- PowerSched can be used to control the power draw of nodes on ARCHER2 and, at the most aggressive setting, could reduce a compute-node's power draw by around 30--40%
- At a coarse level, PowerSched power capping provides a similar control to the frequency capping (already implemented on ARCHER2).
- As with frequency capping, constraining the power draw always increases the runtime but can yield energy efficiency improvements over the uncapped scenario.
- The most aggressive caps have a dramatical impact on both energy efficiency and wall-time (some jobs take nearly twice as long to run and consume 2—3 times as much energy).
- Use of PowerSched is not likely to obviate the need to power down ARCHER2 nodes during hot-weather conditions but could be a useful tool in warm-weather conditions.

A Request for Change is, at the time of writing, being considered for a full-system trial of PowerSched as an alternative to frequency capping. This would allow the ARCHER2 management to evaluate PowerSched as an alternative to frequency capping (for example, comparing effectiveness, user experience, and energy use).

Updates to the ARCHER2 GPU Development System

The previously reported work to update the development tools and drivers on the ARCHER2 GPU Development System was completed during the period and a new image has been rolled out featuring AMD ROCm Version 6.3 tooling and libraries (a significant update on the previous version).

Initial feedback from users confirms that code performance and reliability has been improved by the update, and that this makes the GPU Development System a viable platform on which to test and develop contemporary AMD GPU software. Of particular note, the new image was used for the CCP9/ARCHER2 GPU acceleration hackathon, which is discussed under Training.

Exit Planning

The ARCHER2 CSE Service is due to end on 22nd November 2026. In preparation for this, the CSE team has begun exit planning, undertaking or contributing to the following activities:

- An area of the ARCHER2 documentation has been created (<https://docs.archer2.ac.uk/end-of-service-2026/>), publicising the end of service and providing guidance for users on, for example, curating their long-term data and what to expect beyond the end of ARCHER2.
- A schedule of regular status meetings has been agreed with EPSRC and NERC.
- HPE and EPCC have shared their high-level exit plans.

Within CSE, function leads have drafted exit plans which will be finalised in the coming weeks and then actioned.

Performance Report

This is the performance report for the ARCHER2 CSE Service for the Reporting Period. The metrics were specified by EPSRC in Schedule 2.2 of ARCHER2 CSE Service Contract, as follows.

CSE Query Metrics

- **ARCHER2_CSE_Level1 (MTR):** The Median Time to Resolution, as measured by Working Days (WDs), of all CSE queries falling within Level 1 resolved by the Contractor in the Reporting Period. *MTR applicable to OY5: Service Threshold: >4 WD; Operating Service Level: >1 WD, ≤2 WD.*
- **ARCHER2_CSE_Level2 (MTR):** The Median Time to Resolution, as measured by Working Days (WD), of all CSE queries falling within Level 2 resolved by the Contractor in the Reporting Period. *MTR applicable to OY5: Service Threshold: >25 Working Days (WD); Operating Service Level: >10 WD, ≤15 WD.*
- **ARCHER2_CSE_Level3 (MTR):** The Median Time to Resolution, as measured by Working Days (WD), of all CSE queries falling within Level 3 resolved by the Contractor in the Reporting Period. *MTR applicable to OY5: Service Threshold: >55 Working Days (WD); Operating Service Level: >25 WD, ≤35 WD.*
- **ARCHER2_CSE_TA (%):** The percentage of the total number of Technical Assessments (TAs) assigned to the Contractor in the Reporting Period completed prior to the commencement of the applicable TA Target Completion Date after the assignment of such Technical Assessment to the Contractor. *TA Target Completion Date in OY5: 6 WD; Service Threshold: <90.00%; Operating Service Level: 95.00-97.49%.*
- **Initial Response to Queries (%):** The percentage of the total number of CSE queries assigned to the Contractor in the Reporting Period responded to within 3 Working Hours. *Service Threshold: <96.00%; Operating Service Level: 98.00 – 98.99%.*
- **Query User Satisfaction (%):** The percentage of the total number of query satisfaction surveys completed in each Reporting Period, rating the quality of the resolution of Queries by the Contractor as “Good”, “Very Good” or “Excellent”. *Operating Service Level: 82.00 – 87.99%.*

- **Training User Satisfaction (%):** The percentage of all training satisfaction surveys completed in each Service Period, rating the Contractor as “Good”, “Very Good” or “Excellent”. *Operating Service Level: 88.00%-92.99%.*

Metric	Apr 2026		May 2026		Jun2026		Q2 2026	
	Perf	Points	Perf	Points	Perf	Points	Perf	Points
ARCHER2_CSE_Level1 (MTR)	0.1WD	-2	0.1WD	-2	0.1WD	-2	0.1WD	-6
ARCHER2_CSE_Level2 (MTR)	0.4WD	-2	0.9WD	-2	0.2WD	-2	0.5WD	-6
ARCHER2_CSE_Level3 (MTR)	0WD	-2	0	-2	0	-2	0WD	-2
ARCHER2_CSE_TA (%)	100%	-3	100%	-3	100%	-3	100%	-3
Initial Response to Queries (%)	98.2%	-1	100%	-1	100%	-1	99.4%	-2
Query User Satisfaction (%)	100%	-2	100%	-2	100%	-2	100%	-6
Training Satisfaction (%)	100%	-3	100%	-3	100%	-3	100%	-3
Total		-10		-9		-9		-28

Table 1: CSE Service Performance during April—June 2026

28 query feedback responses were received on query resolution in the Reporting Period. 100% of responses had a score of “Good”, “Very Good” or “Excellent”.

CSE Queries

This section provides details on ARCHER2 CSE queries during the Reporting Period.

CSE Query Statistics

The metrics were specified by EPSRC in Schedule 2.2 of ARCHER2 CSE Service Contract.

- **Assigned:** The number of CSE queries assigned to the Contractor within each query resolution category in the Reporting Period.
- **Resolved:** The number of CSE queries resolved by the Contractor within each query resolution category in the Reporting Period.
- **Backlog:** The number of CSE queries assigned to the Contractor that remained unsolved within each query resolution category in the Reporting Period.
- **Correspondence:** The average number of pieces of correspondence generated for CSE queries in each query resolution category.
- **First Response:** The average time taken for the Contractor to first respond to the Originator of the CSE query.

Apr 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
ARCHER2_CSE_Level1	32	32	0	3	0.1h
ARCHER2_CSE_Level2	25	22	20	12	0.7h
ARCHER2_CSE_Level3	0	1	1	38	0.2h
ARCHER2_CSE_TA	2	1	1	4	0.2h
May 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
ARCHER2_CSE_Level1	18	18	0	2	0.2h
ARCHER2_CSE_Level2	18	33	5	12	0.2h
ARCHER2_CSE_Level3	0	0	1	0	-
ARCHER2_CSE_TA	4	5	0	7	0.1h
Jun 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
ARCHER2_CSE_Level1	39	39	0	2.051	0.3h
ARCHER2_CSE_Level2	34	20	19	12.7	0.2h
ARCHER2_CSE_Level3	0	0	1	0	-
ARCHER2_CSE_TA	4	4	0	5	0.2h
Q2 2026					
Service level	Assigned	Resolved	Backlog	Correspondence	First Response
ARCHER2_CSE_Level1	89	89	0	2.213	0.2h
ARCHER2_CSE_Level2	77	75	19	12.147	0.3h
ARCHER2_CSE_Level3	0	1	1	38	0.2h
ARCHER2_CSE_TA	10	10	0	6	0.1h

CSE Query Categories

A total of 175 queries were resolved by the ARCHER2 CSE service in the Reporting Period. Resolved CSE queries in the Reporting Period fell into the following categories:

Service level	Category	N. Resolved	% Queries
ARCHER2_CSE_Level1	Courses	89	50.9%
ARCHER2_CSE_Level2	3rd party software	20	11.4%
	Software errors	11	6.3%
	Batch system and queues	9	5.1%
	Software installation	9	5.1%
	Compilers and system software	6	3.4%
	Courses	4	2.3%
	Login, passwords and ssh	4	2.3%
	Storage and compute resources	4	2.3%
	Porting, performance and scaling	3	1.7%
	Data transfer	2	1.1%
	eCSE applications/calls	2	1.1%
	Access to services	1	0.6%
ARCHER2_CSE_Level3	3rd party software	1	0.6%
ARCHER2_CSE_TA	Pump-priming	7	4.0%
	Grant	3	1.7%
Total		175	100.0%

Training

CSE has continued to develop and deliver a training programme for the ARCHER2 community. During the second quarter of 2026, the CSE service has provided a total of 22.5 days of training, scheduled as follows:

Dates	Course	Location	Days	Attend
14-15 Apr	Profiling and Optimisation Workshop	Online	2	13
15 Apr	Discrete element method simulations on national HPC facilities	Online	0.5	11
15-17 Apr	Modern C++ for Computational Scientists	Online	3	16
13-16 Apr	Intermediate Research Software Development	Manchester	3.5	13
22 Apr	An ARCHER2 User's Spack Primer	Online	0.5	16
22-23 Apr	Message Passing programming with MPI	IC, London	2	12
27-28 Apr	Workflows	Online	1	9
29 Apr	Learning (from) protein dynamics	Online	0.5	17
1 May	Message Passing programming with MPI (contd.)	Online	0.5	12
13 May	Shaken, not stirred! Influence of external vibration on phase change	Online	0.5	11
13-15 May	CCP9/ARCHER2 hackathon on directive-based GPU acceleration of Fortran codes	Daresbury	2.5	27
27-29 May	GPU programming with kernels	Online	3	4

Dates	Course	Location	Days	Attend
10 Jun	Digital Oceans: Mapping and Predicting Ireland's Ocean	Online	0.5	24
22, 24 Jun	Containers for Reproducible Research: Introduction to Podman and Singularity	Online	2	5
24 Jun	SEAVEA toolkit: An introduction to verification, validation and uncertainty quantification for computational applications	Online	0.5	20

The most popular event was the CCP9/ARCHER2 GPU acceleration hackathon with 27 participants, while specialist seminars such as Digital Oceans (24 attendees) and the SEAVEA toolkit workshop (20 attendees) also generated strong interest.

On the feedback for online courses, attendees rated a course on a scale of 1–5 (“Very Bad”, “Bad”, “Good”, “Very Good”, and “Excellent”).

The average feedback using this metric was 4.3: that is, better than “Very Good”. Users provided 32 responses: a response rate of 34%. Please note that the feedback from the Containers course has not been included due to its proximity to the date of this report.

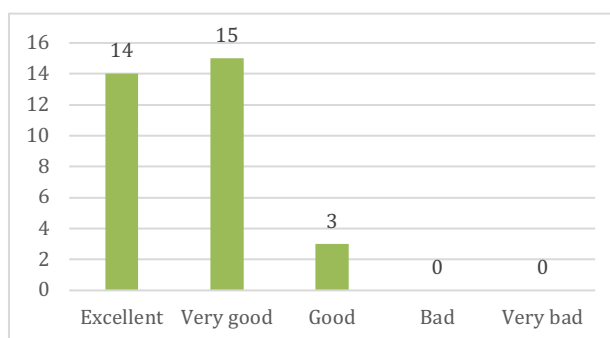


Table 2: Participant feedback for online training courses held during the period.

Embedded CSE Programme (eCSE)

The eCSE team has continued to collect final reports from completed ARCHER2 eCSE projects during the period, and to publish summaries of these on the [ARCHER2 website](#).

At the time of writing, two ARCHER2 eCSE projects are on-going, and all should be finished by the end of the ARCHER2 service. Eleven of the twelve GPU eCSE projects are on-going.

The eCSE team delivered a very successful session on [Research software development funding calls](#) at the [Durham HPC Days](#), during 15th–19th June.

Community Engagement, Outreach, Collaboration and Impact

Benefits Realisation

Blogs

EPCC has published ten ARCHER2 blog articles over the period, ranging from the reduction in a British Geological Survey model time from 6 days to 1 hour, to user experiences of running on capability days.

Community and Outreach Activities

A focus of attention, during the period, has been on the preparations for the Royal Society Summer Science Exhibition (30th June–5th July, London). The ARCHER2 team has been accepted as one of a only 13 exhibitors at this prestigious event. The ARCHER2 team has developed an exhibit to showcase

the value of supercomputing to the public and to consider the environmental cost of running these services and how this cost is being addressed.

Diversity and Inclusivity

During the review period, Women in HPC (WHPC) continued to strengthen its impact through a programme of high-profile events and community engagement activities.

On 29th April 2026, Eleanor Broadway delivered a keynote presentation, "[The Work Behind The Work: Building Community & Careers](#)", at the SSI Collaborations Workshop 2026. Drawing on her experience volunteering with WHPC, the keynote explored how community-building activities have contributed to her own professional development and opportunities, while highlighting the broader value of everyone investing in diversity, inclusion, and community engagement. The talk encouraged attendees to recognise, value, and support the unseen work that enables our community to thrive and benefit from everyone contributing meaningfully. [Read Eleanor's reflections here.](#)

At the Durham HPC Days 2026, on 18 June 2026, WHPC organised a session titled "We know what to do, so why aren't we doing it? Closing the Implementation Gap: Lessons from EuroHPC, UKRI and Beyond." The session was led by EPCC in collaboration with the WHPC Cambridge and East Anglia Chapter, bringing together three speakers for different perspectives on equality, diversity, and inclusion (EDI):

- EDI: Thoughts from EuroHPC
- EDI Challenges in EPSRC Research Infrastructure: Strategic Infrastructure
- CAKE-CoSeC EDI Working Group: EDI for Real-World Workloads

The session had good attendance and engagement with discussion on the practical actions that individuals, leaders, and organisations can take to close the gap between knowing what to do to improve EDI and practical meaningful implementation. Participants explored immediate actions and the wider structural changes that are needed to embed improvements.

At ISC High Performance 2026 (22–26 June 2026), [WHPC delivered its largest and most visible conference programme to date!](#) Building on the renewed partnership between WHPC and the ISC Group, established in 2024, the partnership significantly increased the visibility of WHPC events and provided new opportunities to engage with both existing members and first-time participants. The conference also showcased diverse and underrepresented voices and strengthened the visibility of role models across the international HPC community.

As usual, the programme included the Tech Talks by Women in HPC featuring invited and lightning talks covering both technical topics and career development, Women in HPC Posters within the main conference poster track and the Women in HPC Poster Reception. Also, several new initiatives were introduced in 2026 to further enhance community engagement and inclusion. The introduction of WHPC supported moderators across the programme to increase the visibility of diverse role models, a dedicated community meet-up that attracted over 100 registered attendees and Eleanor Broadway joining the ISC High Performance Steering Committee to provide strategic advice on diversity and inclusion across the conference.

Quality Management, Information Security and Business Continuity

EPCC is delighted to announce that we have passed our ISO external audit with flying colours. This was for three internationally recognised standards:

- ISO 9001 for quality service delivery
- ISO 27001 for information security
- ISO 22301 for business continuity and disaster recovery

The investment in staff time and expense EPCC invests in compliance reflects the importance we give to ensuring we apply best practice in the delivery of the services we run and the safe handling of our user and customer data.